Kostiantyn Pysanyi

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Professional Summary

Computer Vision and Machine Learning Engineer with a strong research foundation and hands-on experience delivering robust end-to-end ML systems. Experienced in developing and deploying deep-learning detection, segmentation, and classification models for real-world imaging applications.

Technical Skills

ML & CV: PyTorch/Lightning, TensorFlow, JAX, Detectron2, MMEngine/MMCV/MMDetection, Albumentations, OpenCV, ITK, pyclesperanto, NumPy, Pandas, CVAT, Transformers/TIMM/Huggingface, Scikit-learn, XGBoost

Experimentation & Tracking: MLflow, Tensorboard, Optuna, DVC, Matplotlib/seaborn/Plotly

Infrastructure & Deployment: Linux, Docker, Kubernetes, Prefect, CI/CD (GitLab, Bitbucket), AWS, Triton/TensorRT

Backend & Dev Tools: Flask, REST, FastAPI, OpenAPI, FHIR, Git, SQL, MQTT

Languages: Python (Professional), C++17, TypeScript (Proficient), Java, MATLAB (Basic)

Experience

Machine Learning & Computer Vision Engineer June 2023 – July 2026
RAYLYTIC GmbH Full-time

- Delivered production-grade **detection**, segmentation, and classification models for X-ray imaging; improved accuracy (mAP, Dice, ROC) by 20-50% over baseline and reduced training time 8-fold
- Built an end-to-end 2D–3D registration pipeline, including mesh processing and alignment, for 3D implant pose estimation from X-rays; improved acceptance rate by 45% and reduced error by 15%; productionized it with **GPU-accelerated Docker** and **Prefect workflow orchestration**
- Implemented an **MLOps** pipeline with dataset versioning via **DVC**, **MLflow** experiment tracking, and automated evaluation
- Migrated model serving to a centralized **Triton** Inference Server–based **REST API**, cutting latency by 50%
- Developed 3D instance segmentation and 3D image registration models for liver cancer research project.

Freelance Computer Vision Engineer Feb 2026 – Mar 2026
Smoothr.de / Sterlix GmbH Contract

- Developed depth-map–based **camera obstruction detection** for real-time retail anti-theft systems

Research Engineer January 2022 – March 2023
Data Science Research Group Part-time

- Developed ML model for bio-signal classification for pain assessment in OP scenarios without patient feedback
- Designed data pipelines and training workflows for a ticket-routing classifier, transforming unstructured text into structured features and streamlining dataset updates and model evaluation for industry partners.

Internships

- ML Thesis Student at RAYLYTIC GmbH – (Oct 2022 – Feb 2023): Developed anatomical landmark localization model
- ML Intern at Prudsys GmbH – (Dec 2021 – Feb 2022): Automated analytic workflows and data preprocessing pipelines
- Autonomous Driving Intern at IAV GmbH – (Sep 2021 – Nov 2021): Investigated inverse reinforcement learning for self-driving
- Student Research Assistant at WHZ – (Jan 2020 – Dec 2021): Developed a semantic segmentation model for automated tissue-density analysis of 3D lung CT scans; developed an autoML library and predicted patient admission to the ICU during COVID.

Education

B.Sc. Data Science with distinction (1.1 / 1.0) Oct 2019 – Apr 2023
University of Applied Sciences Zwickau Zwickau, Germany

M.Sc. Civil Engineering (wGPA: 90.12/100) Sep 2014 – Jun 2020
Odesa State Academy of Civil Engineering and Architecture Odesa, Ukraine

Publications & Awards

EFORT 2026: Validation of a 2D-3D Image Registration Method for TKA Implant Pose Estimation. Pysanyi et al.

DWG 2023: Automatic determination of lumbar segmental lordosis based on an AI algorithm. Pysanyi et al.

1st Place: National Strength of Materials Olympiad (Ukraine, 2018)

Languages

German (C1, TestDaF), English (C2, EF SET Certificate)